Supplementary Material

A coupled whole-plant simulation model with computed carbon input: photoperiod-dependent growth in *Arabidopsis thaliana*

S1. Parameter provenance ledger

This table is the complete machine-readable provenance ledger of the released model, reproduced from `config/parameter_provenance.csv`. It is the full form of the curated physical parameter table given in the main text (Table 2). The tier column states how each value was obtained: T1 measured, T2 derived, T3 inherited from the source model, T4 constrained by physics or chemistry, T5 fitted, T6 unsourced. The ledger holds 124 entries: 4 measured (T1), 55 derived (T2), 27 inherited (T3), 9 constrained (T4), 15 fitted (T5) and 14 unsourced (T6). Every key of the released configuration appears in the ledger, as does every constant hard-coded in the model source, and every ledger value that names a configuration key matches the value the reported runs load; these properties are checked by script rather than by reading.

Table 1: **Table S1.** Complete parameter provenance ledger, generated from `config/parameter_provenance.csv`. Locator gives the equation, section or source line from which the value comes; a dash means no locator is recorded.

Parameter	Value	Unit	Tier	Source	Locator
gas_exchange.leaf_temperature_day	21	degC	T1	Sulpice2014	tools/model_report.c:160, tools/validation_runner.c:628
gas_exchange.leaf_temperature_night	19	degC	T1	Sulpice2014	tools/model_report.c:161, tools/validation_runner.c:629
gas_exchange.specific_leaf_area	0.0055	m2 gFW-1	T1	Tolleter2024	Table 2, reference macroscopic metrics, Col-0 leaf 6 at 21 DAI: leaf a
photo.photoperiod	8	h	T1	Sulpice2014	Methods / Supplemental Table 1 (4, 6, 8, 12 h treatments)
carbohydrates.k_vacuole_out	0.041959	h-1	T2	Seydel2025	Plant Physiol 198:kiaf117, Suppl. Table S5 sheet ST3, control rows, n=
carbohydrates.night_efficiency_starch	0.5	-	T2	–	input.c:31; default.config:30
carbohydrates.starch	0.225	umolC6/gFW	T2	–	input.c:42; default.config:41
carbohydrates.starch_degradation_rate	0		T2	–	–
carbohydrates.starch_night_start	2	umolC6/gFW	T2	–	–
carbohydrates.starch_partition_coeff	0.6	-	T2	–	input.c:25; default.config:24
carbohydrates.sucrose	1.95	umolC6/gFW	T2	–	input.c:41; default.config:40
carbohydrates.sucrose_vacuole	0	umolC6/gFW	T2	–	–
carbohydrates.transport_cost	0		T2	–	–
carbohydrates.uptake_cost	0	umol C6 gFW-1 h-1	T2	Tedone2020	§2.2.8 (n_c*U_N + p_c*U_P); config entry is a slot
gas_exchange.convert_to_photosynthesis_per_gram_per_hour	24000	dimensionless	T2	–	–
gas_exchange.leaf_temperature	25		T2	–	Default configuration value, config/default.config:97. Overridden in e
gas_exchange.light_limited_photosynthesis	10		T2	–	–
gas_exchange.light_limited_rate	10		T2	–	–
gas_exchange.net_photosynthesis	10		T2	–	–
gas_exchange.net_photosynthesis_rate	10		T2	–	–
gas_exchange.respiration_rate	1.5		T2	–	–
gas_exchange.rubisco_limited_photosynthesis	12		T2	–	–
gas_exchange.rubisco_limited_rate	12		T2	–	–
growth.leaf_biomass	0.02	gFW	T2	–	input.c:14; default.config:15
growth.root_biomass	0.02	gFW	T2	–	input.c:15; default.config:16
growth.sucrose_root_allocation	0.5	-	T2	–	input.c:86; default.config:74
growth.total_biomass	0.04	gFW	T2	–	input.c:16; default.config:17
hardcoded.jarvis.gs_max	0.707107	mol m-2 s-1	T2	–	photosynthesis.c:586, maximal unstressed conductance
hardcoded.photosynthesis.ref_specific_leaf_area	0.025	m2 gFW-1	T2	–	–
nutrients.assimilation_cost_nitrogen	0.0460308	umolN/umolC6	T2	Tedone2020	Eq 2.2.16, Appendix A
nutrients.assimilation_cost_phosphorus	0.00460308	umolP/umolC6	T2	Tedone2020	c_sp = c_sn/O, Appendix A (O = 10 umolN/umolP)
nutrients.max_nitrogen	23	umolN/gFW	T2	Tedone2020	Eq 2.2.25 (n_max = D*n_min, D = 4)
nutrients.max_phosphorus	2.3	umolP/gFW	T2	Tedone2020	after Eq 2.2.25 ("p_max = n_max / O") with O = 10 from Eq 2.2.7
nutrients.min_nitrogen	5.75	umolN/gFW	T2	Tedone2020	Eq 2.2.25
nutrients.min_phosphorus	0.575	umolP/gFW	T2	Tedone2020	after Eq 2.2.25 ("p_max = n_max / O") with O = 10 from Eq 2.2.7
nutrients.min_phosphorus_photosynthesis	0.575	umolP/gFW	T2	Tedone2020	after Eq 2.2.25 ("p_max = n_max / O") with O = 10 from Eq 2.2.7
nutrients.nitrogen	8.625	umolN/gFW	T2	–	input.c:43; default.config:42
nutrients.nitrogen_affinity	0.5		T2	–	–
nutrients.nitrogen_cost	0		T2	–	–
nutrients.nitrogen_nutrient_uptake	0		T2	–	–
nutrients.nitrogen_uptake	0		T2	–	–
nutrients.phosphorus	0.8625	umolP/gFW	T2	–	input.c:44; default.config:43
nutrients.phosphorus_affinity	0.5		T2	–	–
nutrients.phosphorus_cost	0		T2	–	–
nutrients.phosphorus_nutrient_uptake	0		T2	–	–
nutrients.phosphorus_uptake	0		T2	–	–
nutrients.pot_nitrogen_uptake	0		T2	–	–
nutrients.pot_phosphorus_uptake	0		T2	–	–
nutrients.stoichiometric_signal	0.5	-	T2	–	input.c:87; default.config:75
photo.light	0		T2	–	–
photo.limitation_of_photosynthetic_rate	0.5		T2	–	–
photo.nitrogen_saturation	0.45		T2	–	–
photo.phosphorus_saturation	0.45		T2	–	–
simulation.days	35		T2	–	–
simulation.dt_hours	0.1		T2	–	–
simulation.gui_enabled	true		T2	–	–
simulation.solver_tolerance	1e-08		T2	–	–
simulation.solver_type	ODE45		T2	–	–
simulation.write_csv	true		T2	–	–
carbohydrates.max_starch_degradation_rate	6	umolC6/(gFW h)	T3	Tedone2020	§2.2.4
carbohydrates.max_sucrose	4.712	umolC6/gFW	T3	Tedone2020	§2.2.4 (Eq 2.2.28)
carbohydrates.min_starch	0.15	umolC6/gFW	T3	Tedone2020	§2.2.4
carbohydrates.min_sucrose	1.3	umolC6/gFW	T3	Tedone2020	§2.2.4
carbohydrates.respiration_frequency	0.79	1/h	T3	Tedone2020	Section 2.2.8 (Costs), "r_m_bar = 0.79 1/h"
carbohydrates.sucrose_loading_frequency	1.98	1/h	T3	Tedone2020	§2.2.8 (Eq 2.2.28)
gas_exchange.michaelis_constant_CO2	404.9	umol mol-1	T3	Bernacchi2001	Rubisco kinetic constants at 25 C, in vivo temperature functions (tran
gas_exchange.michaelis_constant_O2	278.4	mmol mol-1	T3	Bernacchi2001	Rubisco kinetic constants at 25 C: Ko = 278.4 mmol mol-1. Same attribu
growth.leaf_competitive_rate	5e-06	1/(h gFW)	T3	Tedone2020	§2.2.10 (paragraph after Eq 2.2.33): "fixed to the small value 5e-6"
growth.leaf_deathrate	5e-06	1/h	T3	Tedone2020	§2.2.10 (paragraph after Eq 2.2.33): "fixed to the small value 5e-6"
growth.root_competitive_rate	5e-06	1/(h gFW)	T3	Tedone2020	§2.2.10 (paragraph after Eq 2.2.33): "fixed to the small value 5e-6"
growth.root_deathrate	5e-06	1/h	T3	Tedone2020	§2.2.10 (paragraph after Eq 2.2.33): "fixed to the small value 5e-6"
hardcoded.jarvis.T0	25	degC	T3	–	photosynthesis.c:536, tempera-